



Maintenance Schedule

Diesel engine

12V4000G73 / T1238A34-2150

16V4000G73 / T1638A34-2152

Application group 3A

MS50009/04E

12V4000G73
16V4000G73

Table 1: Applicability

Rolls-Royce Solutions refers to Rolls-Royce Solutions GmbH or an affiliated company pursuant to § 15 AktG (German Stock Corporation Act) or a controlled company (joint venture), and Rolls-Royce Solutions Ruhstorf GmbH.

© Copyright Rolls-Royce Solutions

This publication is protected by copyright and may not be used in any way, whether in whole or in part, without the prior written consent of Rolls-Royce Solutions. This particularly applies to its reproduction, distribution, editing, translation, microfilming and storage and/or processing in electronic systems including databases and online services.

All information in this publication was the latest information available at the time of going to print. Rolls-Royce Solutions reserves the right to change, delete or supplement the information provided as and when required.

1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product from the manufacturer Rolls-Royce Solutions.

Maintenance concept

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience and therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

Adherence to the specified maintenance intervals is essential to maintain product safety.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL 1 to QL4.

QL 1: Operational monitoring and maintenance work which does not require disassembly of the product.

QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).

QL3: Maintenance work which requires partial disassembly of the product.

QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the product is to remain out of service for a longer time period, Rolls-Royce Solutions recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at <http://www.mtu-solutions.com>.

1.2 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

1.3 Load profile

Load profile

Power	110%	100%
Corresponding operating time	1%	99%

1.4 Maintenance schedule matrix

0 to 14,500 operating hours

Task	Limit	Operating hours [h]																													
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000	12,500	13,000	13,500	14,000	14,500
Engine																															
W1024	1 w																														
W0500	-	X																													
W0501	-	X																													
W0503	-	X																													
W0505	-	X																													
W0506	-	X																													
W0525	-	X																													
W1009	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1864	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1008	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1247	3 m			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1463	1 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1575	1 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1001	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1241	2 a							X						X						X						X					
W1005	3 a							X						X						X						X					
W1207	18 a			X				X						X						X						X					
W1689	18 a							X						X						X						X					
W1006	18 a													X												X					
W1713	18 a													X												X					
W1011	18 a													X												X					
W1046	18 a																														
W1026	5 a																														
W1812	5 a																														
W1250	6 a																														
W2000	18 a																														
W2001	18 a																														
W2004	18 a																														
W2006	18 a																														
W2007	18 a																														
W2008	18 a																														
W2009	18 a																														
W2010	18 a																														
W2062	18 a																														
W2070	18 a																														
W2110	18 a																														
W1041	18 a																														
W1058	18 a																														
W1251	6 a																														
W1134	18 a																														
W3000	18 a																														
W3001	18 a																														
w = weeks m = months a = years																															

w = weeks
m = months
a = years

Task	Limit	Operating hours [h]																													
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000	12,500	13,000	13,500	14,000	14,500
W3002	18 a																														
W3003	18 a																														
W3004	18 a																														
W3005	18 a																														
W3010	18 a																														
W3018	18 a																														
W3021	18 a																														
W3042	18 a																														
W3083	18 a																														
W3100	18 a																														
W3103	18 a																														
W3108	18 a																														
W3177	18 a																														
W3182	18 a																														
W1124	18 a																														
W1878	18 a																														
W1478	18 a																														
w = weeks m = months a = years																															

15,000 to 29,500 operating hours

Task	Limit	Operating hours [h]																													
		15,000	15,500	16,000	16,500	17,000	17,500	18,000	18,500	19,000	19,500	20,000	20,500	21,000	21,500	22,000	22,500	23,000	23,500	24,000	24,500	25,000	25,500	26,000	26,500	27,000	27,500	28,000	28,500	29,000	29,500
Engine																															
W1024	1 w																														
W0500	-																														
W0501	-																														
W0503	-																														
W0505	-																														
W0506	-																														
W0525	-																														
W1009	2 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1864	2 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1008	2 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1247	3 m	X		X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1463	1 a	X		X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1575	1 a	X		X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1001	2 a	X		X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1241	2 a	X						X						X						X						X					
W1005	3 a	X						X						X						X						X					
W1207	18 a	X						X						X						X						X					
W1689	18 a	X						X						X						X						X					
W1006	18 a							X												X											
W1713	18 a							X												X											
W1011	18 a							X												X											
w = weeks m = months a = years																															

Task	Limit	Operating hours [h]																													
		15,000	15,500	16,000	16,500	17,000	17,500	18,000	18,500	19,000	19,500	20,000	20,500	21,000	21,500	22,000	22,500	23,000	23,500	24,000	24,500	25,000	25,500	26,000	26,500	27,000	27,500	28,000	28,500	29,000	29,500
W1046	18 a	X																													
W1026	5 a	X																													
W1812	5 a	X																													
W1250	6 a	X																													
W2000	18 a	X																													
W2001	18 a	X																													
W2004	18 a	X																													
W2006	18 a	X																													
W2007	18 a	X																													
W2008	18 a	X																													
W2009	18 a	X																													
W2010	18 a	X																													
W2062	18 a	X																													
W2070	18 a	X																													
W2110	18 a	X																													
W1041	18 a	X																													
W1058	18 a	X																													
W1251	6 a																														
W1134	18 a																														
W3000	18 a																														
W3001	18 a																														
W3002	18 a																														
W3003	18 a																														
W3004	18 a																														
W3005	18 a																														
W3010	18 a																														
W3018	18 a																														
W3021	18 a																														
W3042	18 a																														
W3083	18 a																														
W3100	18 a																														
W3103	18 a																														
W3108	18 a																														
W3177	18 a																														
W3182	18 a																														
W1124	18 a																														
W1878	18 a																														
W1478	18 a																														
w = weeks m = months a = years																															

30,000 to 30,000 operating hours

Task	Limit	Operating hours [h]																											
		30,000																											
Engine																													
W1024	1 w																												
W0500	-																												
W0501	-																												
W0503	-																												
W0505	-																												
W0506	-																												
W0525	-																												
W1009	2 a	X																											
W1864	2 a	X																											
W1008	2 a	X																											
W1247	3 m	X																											
W1463	1 a	X																											
W1575	1 a	X																											
W1001	2 a	X																											
W1241	2 a	X																											
W1005	3 a	X																											
W1207	18 a	X																											
W1689	18 a	X																											
W1006	18 a	X																											
W1713	18 a	X																											
W1011	18 a	X																											
W1046	18 a	X																											
W1026	5 a	X																											
W1812	5 a	X																											
W1250	6 a	X																											
W2000	18 a	X																											
W2001	18 a	X																											
W2004	18 a	X																											
W2006	18 a	X																											
W2007	18 a	X																											
W2008	18 a	X																											
W2009	18 a	X																											
W2010	18 a	X																											
W2062	18 a	X																											
W2070	18 a	X																											
W2110	18 a	X																											
W1041	18 a	X																											
W1058	18 a	X																											
W1251	6 a	X																											
W1134	18 a	X																											
W3000	18 a	X																											
W3001	18 a	X																											
W3002	18 a	X																											
W3003	18 a	X																											
w = weeks m = months a = years																													

Task	Limit	Operating hours [h]																													
		30,000																													
W3004	18 a	X																													
W3005	18 a	X																													
W3010	18 a	X																													
W3018	18 a	X																													
W3021	18 a	X																													
W3042	18 a	X																													
W3083	18 a	X																													
W3100	18 a	X																													
W3103	18 a	X																													
W3108	18 a	X																													
W3177	18 a	X																													
W3182	18 a	X																													
W1124	18 a	X																													
W1878	18 a	X																													
W1478	18 a	X																													
w = weeks m = months a = years																															

1.5 Maintenance tasks

Qualification	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
Engine						
QL1	-	1 w	Emergency air shut-off flap	Check function of emergency air shut-off flaps (electrical).	X	W1024
QL1	Daily	-	Engine operation	Check engine oil level.		W0500
				Visually inspect engine for general condition and leaks.		W0501
				Check maintenance indicator of air filter.	X	W0503
				Check relief bores of coolant pump(s).		W0505
				Check for abnormal running noises, exhaust gas color and vibration.		W0506
				Check battery-charging generator for contamination, clean as necessary.	X	W0525
QL1	500	2 a	Centrifugal oil filter	Check layer thickness of oil residue, clean out and replace filter sleeve together with every oil change, at the latest.	X	W1009
QL1	500	2 a	Automatic oil filter	Check and clean oil indicator filter.	X	W1864
QL1	500	2 a	Engine oil filter	Replace engine oil filters each time the engine oil is changed or, at the latest, on expiry of the time limit (given in years).	X	W1008
QL1	1000	3 m	Emergency air shut-off flap	Lubricate emergency air shut-off flaps at lubrication points.	X	W1247
QL1	1000	1 a	Engine mounting	Check general condition of engine mounting (visual inspection).	X	W1463
				Check general condition of trunnion mounts (visual inspection).	X	W1575
QL1	1000	2 a	Fuel filter	Replace fuel filter or fuel filter element.		W1001
QL1	3000	2 a	Belt drive	Check condition of belt drive and replace as necessary; Adjust tension.	X	W1241
QL1	3000	3 a	Air filter	Replace air filters.	X	W1005
QL1	3000	18 a	Valve gear	Check valve clearance, adjust if necessary. IMPORTANT! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul.		W1207
QL1	3000	18 a	Cylinder head	Measure valve protrusion.		W1689
QL1	6000	18 a	Fuel injection valves / injectors	Replace fuel injection valves / injectors.		W1006
QL1	6000	18 a	Engine control unit	Injector: Reset drift compensation parameters (CDC).		W1713
QL1	6000	18 a	Combustion chambers	Inspect combustion chambers using an endoscope.		W1011
QL1	15000	18 a	Crankcase ventilation	Replace filter or filter element.		W1046
QL3	15000	5 a	Engine mounting	Check buffer clearance of resilient mounts, check securing screws for secure seating.	X	W1026
				Check for cracks. Measure height of rubber element.	X	W1812
QL3	15000	6 a	Rubber sleeves	Replace all rubber sleeves.		W1250
QL3	15000	18 a	Component maintenance	Before starting maintenance work, drain coolant and flush coolant circuits.		W2000
				Check rocker arm and valve bridge for wear. Visually inspect swing follower and camshaft running faces with endoscope through push-rod bore.		W2001
				Replace HP fuel sensor.		W2004
				Check engine coolant thermostat and replace thermal actuator.		W2006
				Check charge-air coolant thermostat and replace thermal actuator.		W2007
				Overhaul engine preheating unit.	X	W2008
				Check centrifugal oil filter for wear.	X	W2009
				Overhaul starter.		W2010
				Replace sealing material for all disassembled components.		W2062
				Overhaul charge-air coolant pump.		W2070
				Overhaul engine coolant pump.		W2110

w = weeks
m = months
a = years

Qualification	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
QL3	15000	18 a	Exhaust turbocharger	Replace exhaust turbocharger.		W1041
QL3	15000	18 a	HP fuel pump	Replace HP fuel pump.		W1058
QL3	30000	6 a	Hose lines	Replace all hose lines.		W1251
QL3	30000	18 a	Cylinder head	Overhaul cylinder heads.		W1134
QL4	30000	18 a	Extended component maintenance	Disassemble engine completely. Check engine components according to assembly instructions; repair or replace as necessary.		W3000
				Replace all elastomeric parts and seals.		W3001
				Replace piston rings.		W3002
				Replace conrod bearings.		W3003
				Replace crankshaft bearings.		W3004
				Replace cylinder liners.		W3005
				Replace fuel delivery pump.		W3010
				Replace camshaft bearings and camshaft thrust bearings.		W3018
				Replace pressure relief valve in HP fuel system.		W3021
				Overhaul battery-charging generator.	X	W3042
				Replace wiring harnesses.		W3083
				Replace rubber elements of engine mounts.	X	W3100
				Replace swing followers and swing-follower shafts.		W3103
				Check vibration damper, replace as necessary.		W3108
				Check engine oil pump, replace as necessary.		W3177
				Check tooth flanks of gear train for wear (visual inspection), replace bearing bushings.		W3182
QL4	30000	18 a	Engine mounting	Replace trunnion bearing elements.	X	W1124
QL4	30000	18 a	Torsionally resilient coupling	Replace rubber elements.		W1878
QL4	30000	18 a	Automatic oil filter	Remove and clean filter housing, replace filter candles.	X	W1478
w = weeks m = months a = years						